S Supplementary Material

Definitions

Definitions used throughout the exam, which are for your reference and correspond to the definitions used in the lecture:

State sequence $X := \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_N \end{bmatrix}, \, x_i \in \mathbb{R}^n$

Input sequence $U := \begin{bmatrix} u_0 \\ u_1 \\ \vdots \\ u_{N-1} \end{bmatrix}, \, u_i \in \mathbb{R}^m$

Nominal state sequence $Z := \begin{bmatrix} z_0 \\ z_1 \\ \vdots \\ z_N \end{bmatrix}, \, z_i \in \mathbb{R}^n$

Nominal input sequence $V := \begin{bmatrix} v_0 \\ v_1 \\ \vdots \\ v_{N-1} \end{bmatrix}, \, v_i \in \mathbb{R}^m$

Open interval $(-a, a) := \{x \in \mathbb{R} | -a < x < a\}$

Closed interval $[-a, a] := \{x \in \mathbb{R} | -a \leq x \leq a\}$

1-norm $\|x\|_1 = \|[x_1, \ldots, x_n]^T\|_1 := \sum_i |x_i|$

∞-norm $\|x\|_\infty = \|[x_1, \ldots, x_n]^T\|_\infty := \max_i(|x_i|)$

2-norm $\|x\|_2 = \|[x_1, \ldots, x_n]^T\|_2 := \sqrt{\sum_i x_i^2} = \sqrt{x^T x}$

Weighted 2-norm $\|x\|_Q = \|[x_1, \ldots, x_n]^T\|_Q := \sqrt{x^T Q x}$

Minkowski sum $\mathcal{A}_1 \oplus \mathcal{A}_2 = \{x_1 + x_2 | x_1 \in \mathcal{A}_1, \; x_2 \in \mathcal{A}_2\}$
(scalar case) $[a, b] \oplus [c, d] = [a + c, b + d]$

Pontryagin difference $\mathcal{A}_1 \ominus \mathcal{A}_2 = \{x_1 | x_1 + x_2 \in \mathcal{A}_1 \forall x_2 \in \mathcal{A}_2\}$
(scalar case) $[a, b] \ominus [c, d] = [a - c, b - d]$

Union $\mathcal{A}_1 \cup \mathcal{A}_2 = \{x | x \in \mathcal{A}_1 \text{ or } x \in \mathcal{A}_2\}$

Intersection $\mathcal{A}_1 \cap \mathcal{A}_2 = \{x | x \in \mathcal{A}_1, \, x \in \mathcal{A}_2\}$

Set difference $\mathcal{A}_1 \setminus \mathcal{A}_2 = \{x | x \in \mathcal{A}_1, \, x \notin \mathcal{A}_2\}$

- If not specified otherwise, an MPC control law refers to $\kappa(x(k)) = u_0^*(x(k))$, i.e. the first element of the optimal input sequence computed by solving the associated optimization problem initialized at $x_0 = x(k)$.

S.1 Standard Nominal MPC Problem

Consider linear systems of the form

$$x(k+1) = Ax(k) + Bu(k). \tag{1}$$

The standard linear MPC control law $\kappa(x(k)) = u_0^*(x(k))$ is defined by

$$
\begin{aligned}
P_{S.1}: \qquad J^*(x(k)) = \min_{U} \quad & l_f(x_N) + \sum_{i=0}^{N-1} l(x_i, u_i) \\
\text{s.t.} \quad & x_{i+1} = Ax_i + Bu_i,\ i = 0, \ldots, N-1, && \text{(2a)} \\
& x_i \in \mathcal{X},\ i = 0, \ldots, N-1, && \text{(2b)} \\
& u_i \in \mathcal{U},\ i = 0, \ldots, N-1, && \text{(2c)} \\
& x_N \in \mathcal{X}_f, && \text{(2d)} \\
& x_0 = x(k). && \text{(2e)}
\end{aligned}
$$

Assumption S.1-1 The stage cost $l(x, u)$ is positive definite and only zero at the origin.

Assumption S.1-2

- The terminal set $\mathcal{X}_f$ is a positively invariant set for system (1) under the controller $\kappa_f(x)$:

$$Ax + B\kappa_f(x) \in \mathcal{X}_f, \quad \text{for all } x \in \mathcal{X}_f.$$

- All state and input constraints are satisfied in $\mathcal{X}_f$:

$$\mathcal{X}_f \subseteq \mathcal{X},\ \kappa_f(x) \in \mathcal{U}, \quad \text{for all } x \in \mathcal{X}_f.$$

Assumption S.1-3 The terminal cost is a continuous Lyapunov function in the terminal set $\mathcal{X}_f$ and satisfies:

$$l_f(Ax + B\kappa_f(x)) - l_f(x) \leq -l(x, \kappa_f(x)), \quad \text{for all } x \in \mathcal{X}_f.$$

S.2 Standard Tube-based MPC Problem

Consider uncertain linear systems of the form

$$x(k+1) = Ax(k) + Bu(k) + w(k) \tag{3}$$

with additive disturbances $w(k) \in \mathcal{W}$, where $\mathcal{W}$ is a compact set containing the origin.

The standard tube-based MPC control law $\kappa_{\text{tube}}(x(k)) = K_{\mathcal{E}}(x(k) - z_0^*(x(k))) + v_0^*(x(k))$ is defined by

$$P_{S.2}: \quad \tilde{J}^*(x(k)) = \min_{V} \quad l_f(z_N) + \sum_{i=0}^{N-1} l(z_i, v_i)$$

$$\text{s.t.} \quad z_{i+1} = Az_i + Bv_i, \ i = 0, \dots, N-1, \tag{4a}$$

$$z_i \in \mathcal{X} \ominus \mathcal{E}, \ i = 0, \dots, N-1, \tag{4b}$$

$$v_i \in \mathcal{U} \ominus K_{\mathcal{E}}\mathcal{E}, \ i = 0, \dots, N-1, \tag{4c}$$

$$z_N \in \tilde{\mathcal{X}}_f, \tag{4d}$$

$$x(k) \in z_0 \oplus \mathcal{E}. \tag{4e}$$

Assumption S.2-1 The stage cost $l(z, v)$ is positive definite and only zero at the origin.

Assumption S.2-2 The set $\mathcal{E}$ is a robust invariant set for system (3) under $u(k) = K_{\mathcal{E}}x(k)$.

Assumption S.2-3

- The terminal set $\tilde{\mathcal{X}}_f$ is a positively invariant set for the nominal system $z(k+1) = Az(k) + Bv(k)$ under the controller $\tilde{\kappa}_f(z(k))$:

  $$Az + B\tilde{\kappa}_f(z) \in \tilde{\mathcal{X}}_f, \quad \text{for all } z \in \tilde{\mathcal{X}}_f.$$

  All tightened state and input constraints are satisfied in $\tilde{\mathcal{X}}_f$:

  $$\tilde{\mathcal{X}}_f \subseteq \mathcal{X} \ominus \mathcal{E}, \ \tilde{\kappa}_f(z) \in \mathcal{U} \ominus K_{\mathcal{E}}\mathcal{E}, \quad \text{for all } z \in \tilde{\mathcal{X}}_f.$$

- The terminal cost is a continuous Lyapunov function in the terminal set $\tilde{\mathcal{X}}_f$ and satisfies:

  $$l_f(Az + B\tilde{\kappa}_f(z)) - l_f(z) \leq -l(z, \tilde{\kappa}_f(z)), \quad \text{for all } z \in \tilde{\mathcal{X}}_f.$$